Progress meeting

# Optimisation-driven quantum circuit reduction

## Reproduction, exact Clifford ground truth, and a general hardware-aware optimiser

**Arnav Kapoor**

IISER Bhopal

Study of B. Rosenhahn, T. J. Osborne, C. Hirche, *Optimization driven quantum circuit reduction*, New J. Phys. **27**, 104509 (2025), arXiv:2502.14715

# The story in one slide

**What I was asked to do**

- Understand the paper, re-run every figure and table.
- Find its weaknesses.
- Extend it.

**What I found**

- It reproduces, but with qualifications.
- Every gate it uses is **Clifford**, so exact optima are computable.
- Against them, the method is **3–4.5**$\times$ from optimal.

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- Against them, the method is **3–4.5**$\times$ from optimal.

**What I built**

- Theory of local optimality, parity and random benchmarks.
- A **general optimiser** for real native gates, connectivity and calibrated noise, with exact, state-aware or approximate semantics.

**Where it stands**

- Proven optimum on 35/40 of the paper's 3-qubit circuits.
- On Qiskit L3 (58 algorithms) it never adds 2q gates and removes 13 % of gates on Heron.
- BQSKit needs fewer 2q gates but is approximate and slower.
- Paper drafted, every output verified.

# Roadmap

1. **The paper.** Database-driven local rewriting
2. **Reproduction.** What holds and what needs qualifying
3. **Ground truth.** Every gate is Clifford $\Rightarrow$ exact optima
4. **Theory.** Exactness, parity, mixing, lower bounds
5. **A general optimiser.** No Clifford crutch, real hardware
6. **Results.** Paper benchmarks, standard suites, BQSKit, noisy simulation
7. **Wrap-up.** Lessons, caveats, critique, next steps

Part I

# The paper

Rosenhahn, Osborne & Hirche shorten circuits by replacing short windows with shorter equivalents from a precomputed database.

# Why shorten circuits at all?

- On NISQ hardware every gate costs fidelity. Roughly,

$$\Pr[\text{success}] \approx \prod_{\text{gates } g} \big(1 - e(g)\big),$$

  with two-qubit errors $\sim 10^{-3}$–$10^{-2}$, 10–100$\times$ single-qubit ones.

- Compiling to a device's **native alphabet** leaves redundancy. One textbook gate becomes several native ones, and neighbours stop cancelling.

- The goal is an equivalent circuit with fewer (or cheaper) native gates.

Figure from arXiv:2502.14715, Fig. 1

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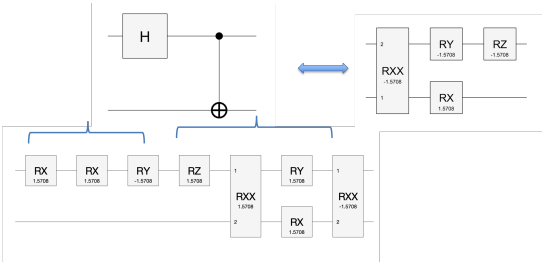
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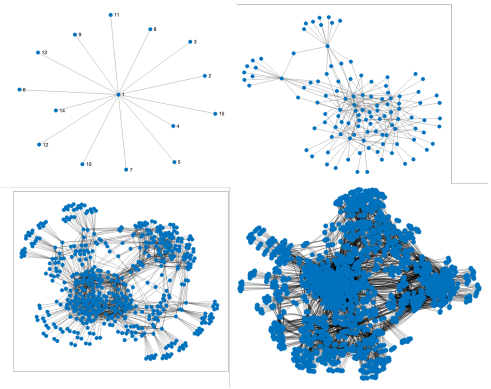



The paper's motivating example, H + CNOT compiled to the ion-trap alphabet (8 gates) and the shortest equivalent native word (4 gates). We confirmed by exhaustive search that no 3-gate version exists.

Figure from arXiv:2502.14715, Fig. 1

# Step 1. The compute graph, a database of shortest circuits

- Fix a **discrete alphabet** on 2–3 qubits
  - ion trap, $R_x, R_y, R_z(\pm\frac{\pi}{2})$, $R_{XX}(\pm\frac{\pi}{2})$
  - IBM-style, $R_x, R_z(\pm\frac{\pi}{2})$, CZ
- Breadth-first search over words. Each node is a unitary (up to global phase), stored with its *shortest* word.
- Depth $d$ = maximal word length enumerated.



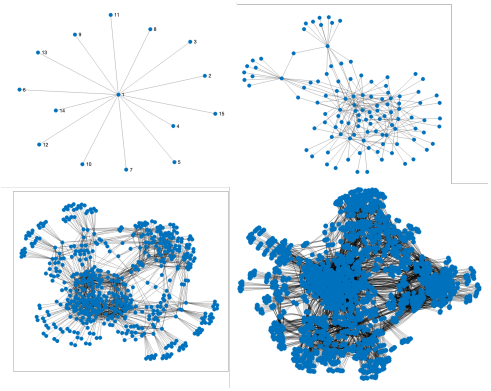
Compute graphs of growing depth (paper, Fig. 2).

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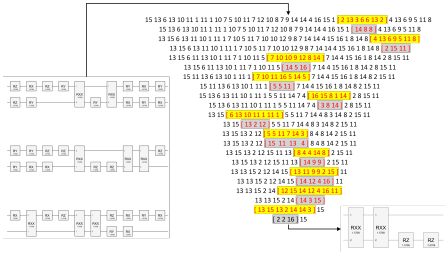
| depth | 1 | 2 | 3 | 4 | 5 |
|---|---|---|---|---|---|
| 2q, 14 ops | 15 | 114 | 584 | 2,024 | 4,512 |
| 3q, 24 ops | 25 | 337 | 3,215 | 23,622 | 137,572 |

Table I node counts. Our rebuild matches the paper exactly.



Compute graphs of growing depth (paper, Fig. 2).

# Step 2. Slide a window, replace, repeat

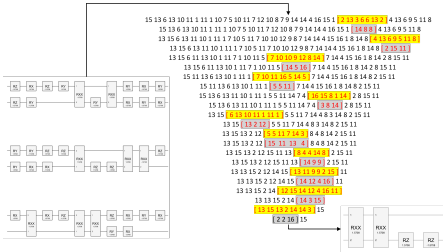


Circuit as a word of tokens, with windows replaced by shorter database words (paper, Fig. 3).

- Pick a window of 2–8 gates on $\leq 3$ wires.
- Look its unitary up and substitute a shorter word if one exists.
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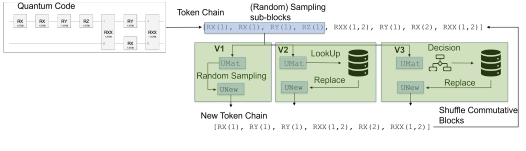
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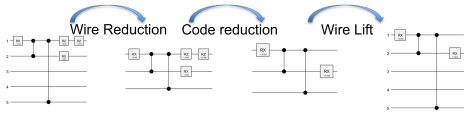
- Pick a window of 2–8 gates on $\leq 3$ wires.
- Look its unitary up and substitute a shorter word if one exists.
- Shuffle commuting gates to expose new windows, and repeat until no progress.




- **V1.** Random sampling for a shorter equivalent word.
- **V2.** Database lookup.
- **V3.** A random forest first predicts whether a window is reducible, and only then is it looked up.

Figures from arXiv:2502.14715, Figs. 3 and 6

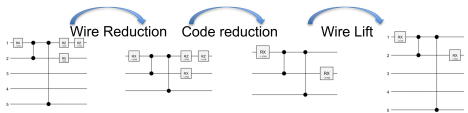
# Step 3. Scaling up, and what the paper claims




Wire reduction, code reduction and wire lift apply a 3-qubit database inside an $n$-qubit circuit.

**Takeaway.** A clean, gate-set-agnostic idea, evaluated against Qiskit only, on random words over a discrete alphabet.

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Wire reduction, code reduction and wire lift apply a 3-qubit database inside an $n$-qubit circuit.

## Claims

| benchmark (random native circuits) | paper | Qiskit (theirs) |
|---|---|---|
| T. III, ion, 3q, 300 gates | 45 | 176 |
| T. IV, IBM-style, 3q, 300 gates | 72 | 153 |
| T. V, IBM-style, 15q, 500 gates | 293 | 313 |
| journal, 4q, 300 gates (ion / IBM) | 111 / 107 | – |
| T. II time (s), V1 / V2 / V3 | 199 / 55 / 38 | |

- Fig. 9 reports better results for reduced circuits on IBM Brisbane and Kyiv.

**Takeaway.** A clean, gate-set-agnostic idea, evaluated against Qiskit only, on random words over a discrete alphabet.

Part II

# Reproduction

Re-running every figure and table with a Python port of the authors' own MATLAB code.

# What I started from, and what I built

**Inputs**

- arXiv PDF + LaTeX source, journal (NJP) PDF.
- Official code `QCOptimDemo.zip`, linked only in the journal's ref. [38] (**V2 only**, MATLAB).
- Not specified in the paper are the database for Tables III–V, the Table II protocol, forest details, the transpiler level and the Qiskit version.

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**Built**

- `qcr/`, a line-by-line Python port (graph builder, V1/V2/V3, wire reduction) with tests.
- An independent exact engine (Clifford tableaux) as a cross-check.
- Every output checked equivalent to its input (unitary error $\sim 10^{-12}$).
- Scripts and raw results for each table and figure, and a reproduction report (PDF).

# Verdict at a glance

| Claim | Status | Detail |
|---|---|---|
| Table I (compute graphs) | **exact** | identical node and edge counts |
| Figs. 1, 4, 5 (examples) | **yes** | same end points, but Fig. 4 needs a different gate pool (parity, Part IV) |
| Tables III/IV, journal 4q | **qualified** | only with the 24-operator, depth-5 database (unstated) |
| Table II, V1 $>$ V2 $>$ V3 | **qualified** | V3 wins only with a tuned forest threshold |
| Table V (15 qubits) | **no longer** | ties a current Qiskit (2.5) |
| Fig. 9 (hardware) | **qualified** | real at Qiskit levels 0/1, disappears at level 3 |

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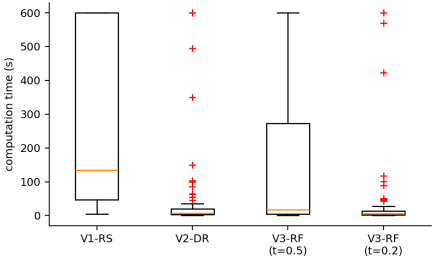
**Takeaway.** The headline claim (beats Qiskit on few-qubit random circuits) holds, but the details around it need qualifying.

# Table II, V1 vs. V2 vs. V3

| variant | mean (s) | median (s) | reached | lookups |
|---|---|---|---|---|
| V1 random sampling | 254.0 | 134.2 | 73 | 51,350 |
| V2 database | 46.4 | 6.5 | 96 | 7,904 |
| V3 forest, $t = 0.5$ | 171.8 | 16.9 | 79 | 915 |
| V3 forest, $t = 0.2$ | 33.7 | 5.3 | 98 | 1,216 |

100 random 5-qubit ion circuits, length 100 $\rightarrow$ target 60, cap 600 s. The paper reports 199 / 55 / 38 s.

Fig. 7 reproduction: 100 random 5-qubit circuits, length 100 → 60

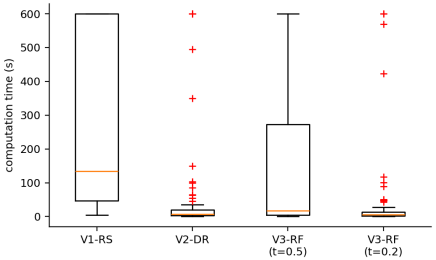


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- V1 ≫ V2 holds, so the database is a real speed-up.
- V3 depends entirely on the forest's **recall**. On held-out windows it flags only **30 %** of reducible ones at the default threshold 0.5 (92 % at 0.2).
- At 0.5 it discards most reductions and is *slower* than V2. The paper reports neither features, forest size, threshold nor accuracy.

**Takeaway.** The "learning" variant only helps after tuning a threshold the paper does not mention, and it saves time on a lookup that need not be slow.

# Tables III–V and the journal statistics against Qiskit

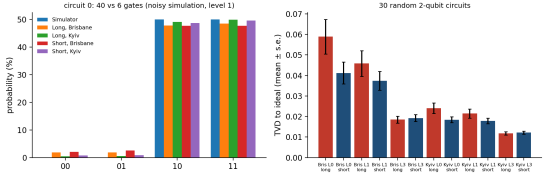
| benchmark | input | Qiskit (paper) | Qiskit 2.5 L3 | paper | our faithful port |
|---|---|---|---|---|---|
| T. III ion, 3q (20 circuits) | 300 | 176 | 127.2 | 45 | **47.1** |
| T. IV IBM-style, 3q (20) | 300 | 153 | 104.0 | 72 | **36.8** |
| journal ion, 4q (30) | 300 | – | 164.9 | 111 | **78.8** |
| journal IBM-style, 4q (30) | 300 | – | 140.9 | 107 | **76.0** |
| T. V IBM-style, 15q (10) | 500 | 313 | 332.1 | 293 | **335.9** |

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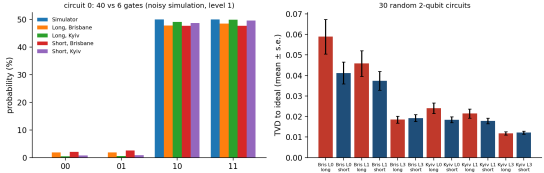
- With few qubits the method clearly beats Qiskit, as claimed, and our port is even better than published.
- But only with the larger database (24 operators, depth 5), which the paper does not state.
- Qiskit 2.5 is much stronger than the paper's Qiskit at 3 qubits (127 vs. 176 and 104 vs. 153).
- At 15 qubits our port reaches 335.9 (paper 293) and only ties Qiskit L3 (332.1, same CZ count). Random 15-qubit gate pairs rarely share wires, so 3-wire windows find little that Qiskit misses.

# Fig. 9, the hardware experiment, in a noisy simulator




| TVD to ideal (30 circuits) | level 0 | level 1 | level 3 |
|---|---|---|---|
| Brisbane, long → reduced | 0.059 → 0.041 | 0.046 → 0.037 | 0.018 vs. 0.019 |
| Kyiv, long → reduced | 0.024 → 0.018 | 0.021 → 0.018 | 0.012 vs. 0.012 |

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- At levels 0/1 the reduced circuit is closer to ideal (22/30 and 20/30 circuits), as in the paper.
- At level 3 Qiskit re-synthesises every 2-qubit block, both versions compile to $\approx 1.1$ CZ, and the effect **vanishes**.
- Plain Qiskit L3 on the *long* circuit beats the reduced circuit at level 1 (0.018 vs. 0.037).
- The paper uses one circuit, one run per chip and no error bars, and does not state the transpiler level.

# Bugs and unstated choices found along the way

**In the released code**

1. `simplereduce` never tests the last adjacent pair, and skips the pair after a deletion. The final circuit in the paper's Fig. 4 still contains an inverse pair (true optimum 2, shown 4).
2. The database excludes the identity, so identity blocks are never deleted by lookup.
3. There are three tolerances, $10^{-5}$ (paper), $10^{-3}$ (optimiser), $10^{-4}$ (graph builder).

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**In the paper**

4. Table I implies a pool of 14/24 operators, but the demo uses 13/21.
5. The database for Tables III–V is unstated, and only the larger one reproduces Table III.
6. Table II states neither the qubit count nor the target length, and the forest is unspecified.
7. For the hardware runs the text says RX, RY, CX but the tables say RX, RZ, CZ, and the transpiler level and shots are unstated.

Part III

# Exact ground truth

The observation the paper misses. Every gate it uses is Clifford, so the search space is a finite group we can enumerate completely.

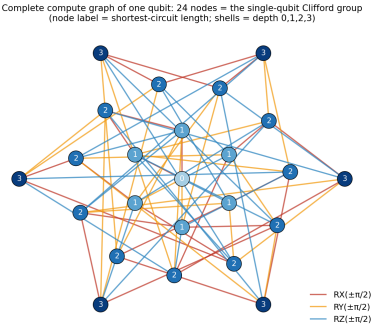
# Key observation, every gate in the paper is Clifford

- $\pm\frac{\pi}{2}$ rotations, $R_{XX}(\pm\frac{\pi}{2})$ and CZ map Paulis to Paulis.
- So every circuit is an element of the finite **Clifford group** $\mathcal{C}_n$ (mod phase), of order

$$2^{n^2+2n}\prod_{j=1}^{n}(4^j-1)$$

(24 for $n=1$, 11,520 for $n=2$, **92,897,280** for $n=3$).

- The paper's "compute graph" is a ball in the **Cayley graph** of this group.




For one qubit the complete compute graph has exactly 24 nodes and diameter 3.

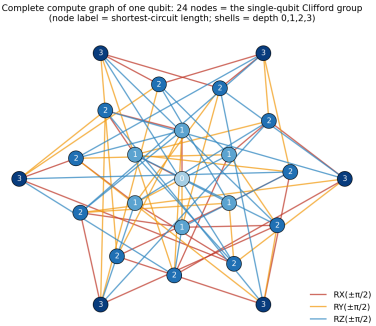
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**Takeaway.** Optimality is no longer a vague question. Each circuit has an exact shortest length, computable by search over one finite graph.

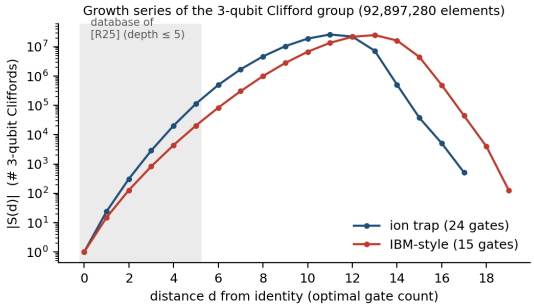
# Enumerating all 92.9 million 3-qubit Cliffords

- A Clifford mod phase is its signed **tableau**, which for $n \leq 3$ packs into one 64-bit word.
- Gates act row-wise through lookup tables, so each BFS level is a vectorised gather. Exact membership, no tolerance.
- Full BFS takes 25 min and $\approx$6 GB per alphabet.

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- Full BFS takes 25 min and ≈6 GB per alphabet.

| alphabet | $n$ | diameter | mean optimal length |
|---|---|---|---|
| ion trap | 2 | 10 | 5.89 |
| IBM-style | 2 | 11 | 6.92 |
| ion trap | 3 | **17** | **10.72** |
| IBM-style | 3 | **19** | **12.34** |




The paper's depth-5 database (shaded) covers 0.15 % of the group, and almost all elements sit at distance 10–14.

# How far from optimal is the paper's method?

| 3 qubits, 300 random gates, 20 circuits | **proven optimum** | paper's method (our port) | Qiskit 2.5 L3 | paper (1 circuit) |
|---|---|---|---|---|
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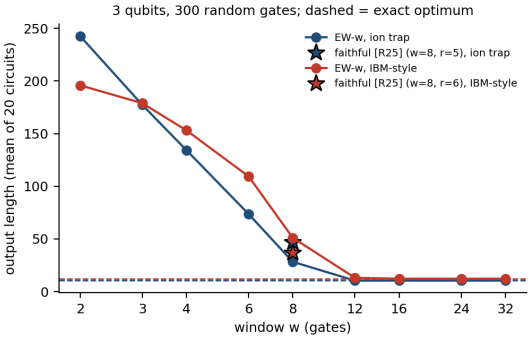
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## The gap

The method's output is **4.5**× (ion trap) and **3.0**× (IBM-style) the optimum, on its own benchmark. Its showcase Fig. 4 output is 2× optimal.

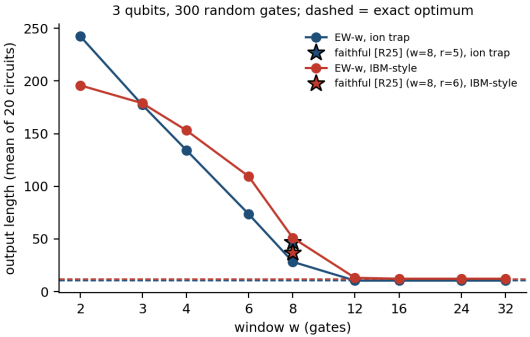
- Our port matches the paper's own single-circuit number (47.1 vs. 45), so this is the method, not the implementation.
- Beating Qiskit by 2.7× looked impressive, yet another 4.5× was available.

# Exact windows close the gap




- **Exact-window optimiser** EW-$w$ replaces every window of $\leq w$ gates on $\leq 3$ wires by its *proven* optimal word (table lookup).

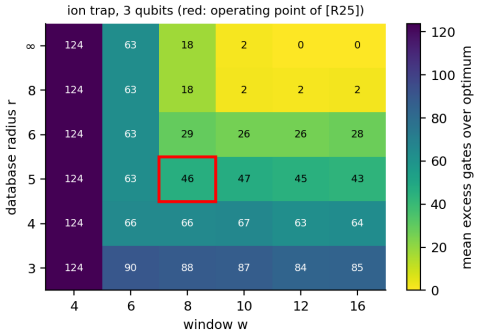
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- **Exact-window optimiser** EW-$w$ replaces every window of $\leq w$ gates on $\leq 3$ wires by its *proven* optimal word (table lookup).
- The transition is sharp. $w \geq 12$ (ion) / $w \geq 16$ (IBM-style) reaches the **proven optimum on all 40 circuits**.
- 0.1–0.2 s per circuit, vs. 51 s / 17 s for the paper's method.
- Why so sharp? Once the circuit is shorter than the window, one more sweep is exact (the *cascade*, Part IV).

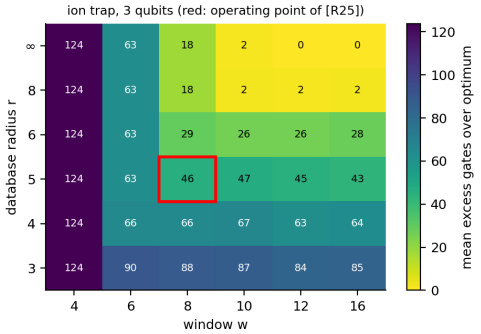
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Mean excess over the optimum (ion trap, 3 qubits) by window $w$ and database radius $r$. The paper's $(8, 5)$ is marked in red.

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Mean excess over the optimum (ion trap, 3 qubits) by window $w$ and database radius $r$. The paper's $(8, 5)$ is marked in red.

- A larger window at $r = 5$ lowers the excess from 46 to 43 and barely helps.
- A larger database at $w = 8$ lowers it from 46 to 18.
- With $w \geq 12$ and $r \geq 8$ the excess is $\leq 2$.

**Takeaway.** The **database radius**, not the window or the search strategy, is the binding constraint. The theory predicts both are needed.

# Four qubits, the journal benchmark

| 4q, 300 gates, 100 circuits | lower bound | CW-sem | CW-24 | contiguous EW-16 | port | published | Qiskit L3 | BQSKit L4 |
|---|---|---|---|---|---|---|---|---|
| ion trap, total | $\geq$11.1 | **45.6** | 56.7 | 110.3 | 78.8 | 111 | 164.9 | 120.6 |
| IBM-style, total | $\geq$12.9 | **49.3** | 59.1 | 130.9 | 76.0 | 107 | 140.9 | 165.2 |
| RXX (ion) / CZ (IBM) | – | – | 27.6 / 27.1 | 44.2 / 47.1 | 33.3 / 32.4 | 43 / 43 | 47.3 / 47.3 | 27.6 / 35.1 |

Port, Qiskit L3 and BQSKit L4 use the first 30 circuits. BQSKit L4 is our run, approximate (error $< 10^{-7}$), all-to-all.

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- Same exact 3-qubit oracle, different windows. Contiguous windows (EW-16) are *no better* than published, **DAG-convex windows** (CW-24) cut the published results by 49 % / 45 %, and adaptive commutation-aware windows (CW-sem) go further.
- **Against BQSKit L4** on the same 30 circuits, CW-24 has about the same RXX count (27.0 vs. 27.6) and fewer CZ (28.1 vs. 35.1, fewer on 25/30), with 0.46× / 0.37× of its total gates, without targeting 2q gates.
- No exact optimum at 4 qubits ($|G| = 1.2 \times 10^{13}$), but a certified lower bound, and the remaining gap is at most $\approx 4\times$.

Port, Qiskit L3 and BQSKit L4 use the first 30 circuits. BQSKit L4 is our run, approximate (error $< 10^{-7}$), all-to-all.

Part IV

# Theory

Local rewriting as rewriting on a Cayley graph. When is it exact, what can it never do, and what do random benchmarks measure?

# The model

- Alphabet $\Gamma$, group $G = \langle \Gamma \rangle$, word length $|g|$, ball $B(r)$, diameter $D$.
- Compute graph of depth $d$ = ball $B(d)$ in $\mathrm{Cay}(G, \Gamma)$, and stored words are geodesics.
- The rewriting system $\mathcal{R}(w, r)$ replaces a factor of length $\leq w$ by an equivalent geodesic of length $\leq r$.

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- The rewriting system $\mathcal{R}(w, r)$ replaces a factor of length $\leq w$ by an equivalent geodesic of length $\leq r$.
- Swapping commuting neighbours = moves in the **trace monoid**.
- Factors of trace-equivalent words = **convex** sets of the circuit DAG.
- So the paper's random commuting shuffles are a *stochastic enumeration of convex windows*, and we enumerate them directly.
- The paper's V2 operates at $(w, r) = (8, 5)$.

# Theorem 1. Local = global above the diameter

## Exactness threshold

Let $G$ be finite with diameter $D$. If $w \geq D + 1$ and $r \geq D$, every $(w, r)$-reduced word is a geodesic.

*Proof idea.* A reduced word longer than $D$ would have a first factor of length $D + 1$, which is not geodesic and fits in the window. So the word has length $\leq D < w$, and then the whole word is one window.

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## Cascade

If at any time the current length $\ell \leq w$, the next sweep returns a geodesic. The relevant window is $w^\star = \min\{w : \ell(w) \leq w\}$.

- Measured $w^\star \approx 12$ (ion) / 16 (IBM-style) for 3 qubits, well below $D + 1 = 18$.
- The paper's $(8, 5)$ is below both thresholds.
- For continuous gate sets $D = \infty$ and no finite window is exact. Hence ground truth must come from the discrete case.

# Theorem 2. Gate-count parity

## Parity

(a) For $n \leq 2$ the Cayley graph is **bipartite**, so every equivalent rewrite removes an *even* number of gates.
(b) For $n \geq 3$, with any generating set containing square roots of each $X_i$, $Z_i$, it is **never** bipartite.

*Proof idea.* Bipartite $\iff$ a sign character $\chi$ with $\chi(\Gamma) = -1$. For $n \geq 3$, $\chi$ kills the Paulis and factors through $\mathrm{Sp}(2n, 2)$, which is simple, so $\chi \equiv 1$.

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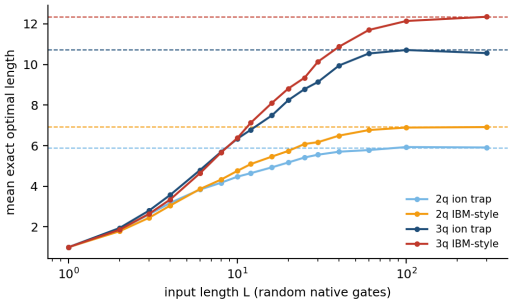
- Exhaustive search at 3 qubits finds a shortest odd relation of length $\rho = 11$ (ion) / 9 (IBM-style). Changing parity needs a window of $\geq 6$ gates, and smaller windows can never reach an optimum of the other parity.
- The paper's Fig. 4 shows a $6 \to 3$ replacement on two qubits, which is **impossible** with the stated gate set.

# Theorem 3. Random-circuit benchmarks measure a constant

## Benchmarks mix

A random word of length $L$ is within $\frac{1}{2}\sqrt{|G|}\,\beta^L$ (total variation) of a uniform group element. So $\mathbb{E}[\text{optimal length}] \to$ the mean distance in $G$.

For 2 qubits (ion trap) $\beta = 0.937$ (sparse diagonalisation).

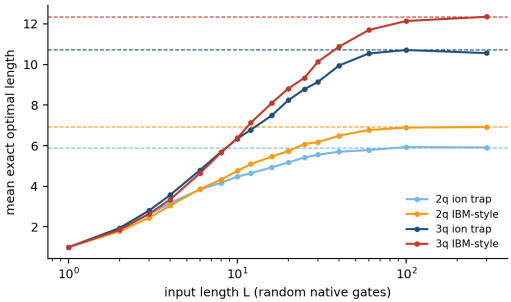


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**Takeaway.** "Reduced from 300 to 45 gates" mostly measures the input length. Beyond $L \approx 100$, every random 3-qubit ion circuit has optimum $\approx 10.7$ on average, whatever $L$ is.

# Lower bounds for every method (4 qubits)

## Counting bound

$\Pr[|g| \leq d] \leq |B(d)|/|G|$, and spheres are submultiplicative, $|S(a+b)| \leq |S(a)||S(b)|$. Exact spheres up to a small depth bound $\mathbb{E}|g|$ from below for any optimiser.

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- Check at $n = 3$, bound 7.9 vs. truth 10.72.
- $n = 4$ ($1.2 \times 10^{13}$ elements), spheres to depth 6 give $\mathbb{E}|g| \ge$ **11.1** (ion) / **12.9** (IBM-style).
- The best achieved is 45.6 / 49.3. Closing that gap is an open problem the framework makes precise.

Part V

# A general optimiser

Drop the Clifford crutch and add continuous angles, connectivity, calibrated noise, and a precise notion of “equivalent”.

# Why go beyond Clifford?

**The Clifford results don't transfer**

- Real compilers emit **arbitrary angles** ($R_z(\theta)$, MS($\theta$)), so there is no finite group and no table.
- Real chips have **connectivity** (heavy-hex) and per-gate **calibrated errors**.
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**Design decisions**

- **No** Clifford assumption in the method. Clifford tables are used only to *evaluate*.
- Devices are IBM Heron (FakeTorino with $R_z(\theta)$, $\sqrt{X}$, $X$, CZ, heavy-hex, calibrated) and a trapped-ion model ($R_{x,y,z}(\theta)$, $R_{XX}(\theta)$, all-to-all).
- Semantics are exact incl. phase, state/measurement-aware, or approximate with tracked error.
- The objective is chosen by noisy simulation, not assumed.

# One idea, equivalence = group action and key = canonical orbit

| notion | $U \sim V$ iff | canonical database key |
|---|---|---|
| exact | $V = U$ | raw matrix |
| up to phase (the paper) | $V = e^{i\varphi}U$ | phase-normalised matrix |
| input-aware (at start) | $V\|0\rangle = e^{i\varphi}U\|0\rangle$ | phase-normalised $U\|0\rangle$ |
| output-aware (before readout) | $V = DU$, $D$ diagonal | rows phase-normalised |
| symmetry | $V = PUP^\dagger$, $P$ a chip symmetry | minimum over the orbit |
| approximate | $\|U - V\| \leq \varepsilon$ | exact key + verified distance |

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## One engine, many semantics

The database stores 64-bit hashes of canonical forms, so one lookup finds the whole class. New semantics = new key function. Every hit is re-verified numerically.

## Proposition 1 (state- and measurement-aware rewriting)

The program prepares $|0^n\rangle$, runs the circuit and measures all qubits. Let $S$ be a convex window on wires $W$. If (a) nothing precedes $S$ on $W$ and $V|0_W\rangle = e^{i\varphi}U|0_W\rangle$, or (b) nothing follows $S$ on $W$ and $V = DU$ with $D$ diagonal, then replacing $U$ by $V$ leaves the output distribution unchanged, and so does any sequence of such rewrites.

*Why.* In (a) the input is a product state, and in (b) $|\langle x|D\psi\rangle|^2 = |\langle x|\psi\rangle|^2$. The invariant is the output distribution.

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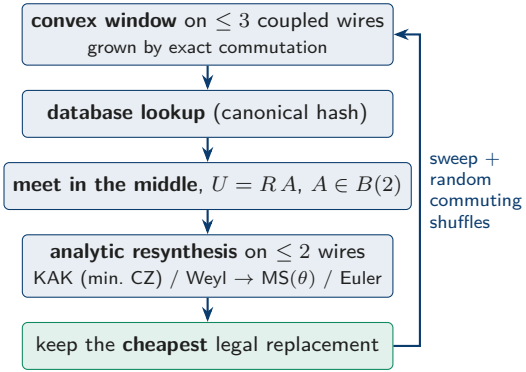
*Why.* In (a) the input is a product state, and in (b) $|\langle x|D\psi\rangle|^2 = |\langle x|\psi\rangle|^2$. The invariant is the output distribution.

## Proposition 2 (approximate rewriting)

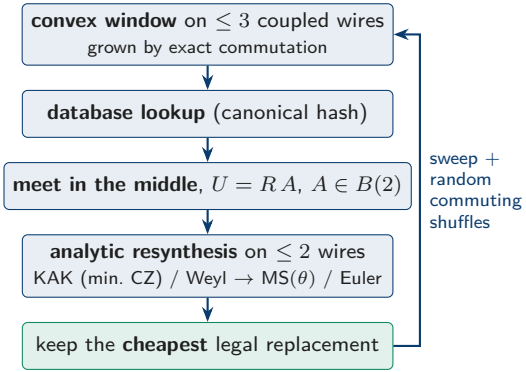
If rewrite $i$ has operator-norm error $\varepsilon_i$, the final circuit is within $\sum_i \varepsilon_i$ of the original (including phase in exact mode).

*Why.* Telescoping, and $\|A(X-Y)B\| = \|X-Y\|$ for unitary $A, B$. The optimiser reports this bound with every circuit.

# The engine


**convex window** on $\leq 3$ coupled wires
grown by exact commutation
**database lookup** (canonical hash)
**meet in the middle**, $U = R\,A$, $A \in B(2)$
**analytic resynthesis** on $\leq 2$ wires
KAK (min. CZ) / Weyl $\rightarrow$ MS($\theta$) / Euler
keep the **cheapest** legal replacement
sweep + random commuting shuffles

# The engine




| device, topology | notion | classes | depth |
|---|---|---|---|
| Heron, line | phase | 6.31 M | 9 |
| | output-aware | 0.34 M | 10 |
| | input-aware | 1.41 M | 14 |
| ion, triangle | phase | 28.4 M | 6 |
| | output-aware | 7.36 M | 6 |
| | input-aware | 31.6 M | 7 |
| paper ion alphabet | phase | 2.93 M | 9 |
| paper IBM alphabet | phase | 1.84 M | 10 |

Grid angles $\pm\frac{\pi}{4}, \pm\frac{\pi}{2}, \pi$ give 23 (Heron) / 57 (ion) local gates. Builds take 18 s to 38 min. Deepest level stored as keys only.

# Global phase and cost, the details that matter

**Exact including global phase**

- Inside windows, match up to phase and record each phase.
- Native gates reach only a *discrete* set of phases, multiples of $\pi/4$ (Heron) and $\pi/2$ (ion). Snap, or reject.
- Repair once at the end with the cheapest gadget. On ions absorb $\pm i$ into an existing MS gate, $R_{XX}(\theta \pm \pi)\, R_x(\pi)^{\otimes 2} = \pm i\, R_{XX}(\theta)$.
- Never return a costlier circuit than the input.

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**Three candidate objectives**

$$c_{\text{count}} = 1, \qquad c_{w2} = \begin{cases} 10 & \text{2q} \\ 1 & \text{1q} \end{cases}$$

$$c_{\text{cal}}(g) = -\log\bigl(1 - e(g)\bigr)$$

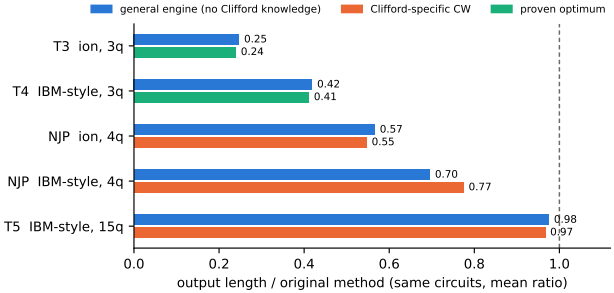
- $e^{-c_{\text{cal}}}$ = estimated success probability (ESP), so minimising $c_{\text{cal}}$ maximises ESP.
- $e(g)$ from FakeTorino calibration per qubit / pair, the ion model uses $3 \cdot 10^{-5}$ / $10^{-3}$, and $R_z$ is virtual.
- Which one wins is decided by noisy simulation (Part VI).

Part VI

# Results

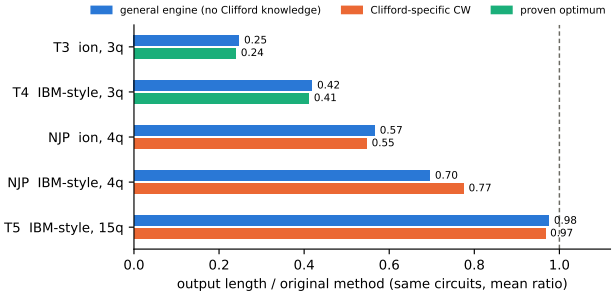
First the paper's own benchmarks against ground truth, then 60 standard circuits on hardware models, against Qiskit, TKET and BQSKit.

# Result 1. The paper's benchmarks, without Clifford knowledge




Output length / paper's method, same circuits and seeds, paper semantics (up to phase, gate count, $\pm\frac{\pi}{2}$ alphabets).

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Output length / paper's method, same circuits and seeds, paper semantics (up to phase, gate count, $\pm\frac{\pi}{2}$ alphabets).

- **3 qubits.** Proven optimum on **17/20** (ion) and **18/20** (IBM-style), mean 10.8 vs. optimum 10.4.
- **4 qubits.** On par with the Clifford-specific optimiser (46.1 vs. 45.6 and 51.4 vs. 49.3).
- **15 qubits.** All methods within 4 %.
- 3–9 s per circuit, and all 250 outputs verified and native.

**Takeaway.** Knowing the group structure is not what makes the difference. Large databases and convex windows are.

**Circuits** (60 per device, 2–12 qubits, $\leq$1,008 gates)

- Nam/VOQC arithmetic (Toffoli/Barenco ladders, adders, modular multiplication, $\mathrm{GF}(2^4)$ multiplier, product formulas, QFT8)
- QASMBench small/medium (4 with classical feed-forward and 4 unparsable skipped)
- QFT, QAOA, Heisenberg Trotter, EfficientSU2, Steane encoder
- Random non-Clifford circuits

# Result 2. Standard suites on hardware models, setup

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**Pipelines** (same native set, coupling map, seed)

- Qiskit transpile, levels 1 and 3
- TKET `FullPeepholeOptimise` on L1 + rebase
- BQSKit 1.2.1, levels 2 and 3 on L1 (approximate resynthesis)
- ours on L1 output and **ours on L3 output**

**Metrics.** 2q count, total, depth, ESP and noisy-simulation TVD ($\leq 8$ active qubits). Every output is verified, native and coupled.

# Result 2. Relative to Qiskit level 3

| | **IBM Heron** (58 circuits) | | | **trapped ion** (58 circuits) | |
|---|---|---|---|---|---|
| **pipeline** | 2q gates | all gates | ESP | 2q gates | all gates |
| TKET | ×1.02 | ×1.71 | ×1.16 | ×1.27 | ×1.79 |
| ours, exact, from Qiskit L1 | ×1.00 | **×0.84** | **×1.21** | **×0.95** | **×0.76** |
| ours, exact, from Qiskit L3 | **×0.98** | **×0.87** | ×1.03 | ×1.00 | ×1.00 |
| ours, approximate ($\varepsilon = 10^{-3}$), from L1 | ×1.01 | **×0.84** | **×1.21** | **×0.95** | **×0.75** |
| ours, state/measurement-aware | **×0.77** | **×0.61** | **×1.28** | **×0.62** | **×0.50** |
| BQSKit L2 on L1 (approximate) | ×0.89 | ×1.26 | **×1.21** | ×0.96 | **×0.74** |
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Ratios of summed counts, and the geometric mean of per-circuit ratios for ESP. Nam pf1_10/pf2_10 are excluded because Qiskit L3 drops their $10^{-7}$ rad rotations (approximate, not exact).

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- **Stacked on L3** ours never adds 2q gates (Heron 9 better and 49 equal, ion 1 and 57) and never lowers ESP.
- TKET keeps 2q counts close to L3 but needs 1.7–1.8× the gates after rebasing.
- BQSKit L3 has the fewest 2q gates, with approximate outputs. Ours on top of it gives the best ESP (next slide).
- Ion ESPs differ by $< 1\,\%$ across pipelines (MS error $10^{-3}$), so they are omitted.

Ratios of summed counts, and the geometric mean of per-circuit ratios for ESP. Nam pf1_10/pf2_10 are excluded because Qiskit L3 drops their $10^{-7}$ rad rotations (approximate, not exact).

# Result 3. BQSKit, the strongest baseline

**Summed over 58 circuits (2q / all gates)**

| | Heron | ion |
|---|---|---|
| Qiskit L3 | 2,889 / 15,404 | 1,378 / 7,425 |
| ours from L1 | 2,902 / **12,908** | 1,305 / 5,608 |
| ours from L3 | 2,830 / 13,405 | 1,376 / 7,420 |
| BQSKit L3 | **2,446** / 18,510 | 1,302 / 5,250 |
| ours on BQSKit L3* | 2,448 / 15,630 | **1,298** / **5,202** |

Same Qiskit L1 input, native gates and couplers. BQSKit 1.2.1 at default settings, output mapped back to the physical qubits. *On a separate BQSKit L3 run (2,459 / 18,601 and 1,301 / 5,276 before our pass).

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- **Heron.** Fewer 2q gates than ours on 39/58 circuits (more on 1), about the same ESP ($\times 1.22$ vs. $\times 1.21$), but $1.38\times$ the gates of ours on L3.
- **Ion.** Fewer gates than all pipelines but our state-aware mode, and a higher ESP than ours on L3 on 45/58.
- **The price.** Approximate (median error $2 \cdot 10^{-8}$, max $1.4 \cdot 10^{-4}$), ours exact with phase. 98 s per Heron circuit on 8 workers, ours 24 s on one.
- **Combined.** Ours on BQSKit's output never adds a 2q gate or lowers ESP, removes 16 % of the Heron gates (ion 1 %) and raises ESP on 54/58, in 34 s.

**Takeaway.** Resynthesis finds cheaper entangling structure. Exact rewriting keeps the unitary, removes single-qubit gates, and stacks on top.

# Where do the gains come from? (honest accounting)

**Stacked on L3, summed over 58 circuits**

| | Qiskit L3 | + ours |
|---|---|---|
| Heron $\sqrt{X}/X$ pulses | 7,218 | 6,139 |
| Heron $R_z$ (virtual) | 5,297 | 4,436 |
| Heron CZ | 2,889 | 2,830 |
| ion $R_x/R_y$ pulses | 3,824 | 2,422 |
| ion $R_z$ (virtual) | 2,223 | **3,622** |
| ion MS | 1,378 | 1,376 |

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- On Heron the savings are split between pulses and $R_z$, plus a few CZ.
- On ions the calibrated objective *trades* pulses for free $R_z$. The total rises on 28/58 circuits, by design. "Never worse" holds for the objective, 2q count and ESP, **not** for the total.
- ESP $\times 1.21$ from L1 decomposes as $\times 1.11$ (Qiskit's L1 layout already beats its L3 layout) $\times$ $\times 1.09$ (our rewriting). We never change the layout.

# Which objective? Let noisy simulation decide

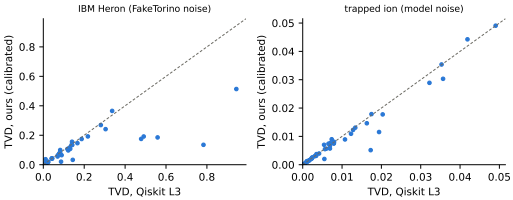
| pipeline / objective | mean TVD (49 circuits) Heron | ion |
|---|---|---|
| Qiskit L1 | 0.115 | 0.0157 |
| Qiskit L3 | 0.137 | 0.0091 |
| ours, total count | 0.096 | 0.0092 |
| ours, 2q-weighted | **0.088** | **0.0084** |
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Exact density-matrix probabilities (no shot noise). Heron uses FakeTorino gate errors of the actual qubits, ion a depolarising model.

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- Calibrated beats L3 on **39/49** (Heron) and 29/49 (ion) circuits.
- Calibrated $\approx$ 2q-weighted $>$ count $\Rightarrow$ calibrated is the default, and 2q-weighted is a good proxy without calibration data.

# The other semantics, state-aware and approximate

## State/measurement-aware

- Largest gains, with 2q gates $\times 0.77$ (Heron) / $\times 0.62$ (ion) relative to L3.
- Output distributions agree to $6 \cdot 10^{-12}$.
- Valid only for a complete program, with $|0 \cdots 0\rangle$ in and full measurement out.

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Caveat. 5 (Heron) / 7 (ion) circuits are reversible logic on $|0 \cdots 0\rangle$ with a fixed basis-state output, and they reduce to the empty circuit. Read this mode per circuit, not as an average.

## Approximate ($\varepsilon = 10^{-3}$ per replacement)

- Almost no gain over exact on these suites, which have few near-identity blocks.
- Measured error always inside the tracked bound (max $6.4 \cdot 10^{-4}$).
- The exception is Nam pf2_10 (rotations $\sim 10^{-7}$ rad), 1,088 $\rightarrow$ 529 gates with error $1.3 \cdot 10^{-4} \leq$ bound $7.1 \cdot 10^{-4}$.

# Every number on these slides is verified

- **Exact mode.** $\|(U - V)\psi\|$ on random states (statevector on the active qubits) is $\leq 8.2 \cdot 10^{-11}$.
- **State-aware.** The TVD of output distributions is $\leq 6 \cdot 10^{-12}$.
- **Approximate.** Measured error $\leq$ tracked bound in all runs.
- **Legality.** Native gates only, and 2q gates only on working couplers.

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- **Coverage.** 250/250 paper-benchmark outputs and 720/720 hardware-suite outputs. BQSKit outputs are checked up to phase and qubit permutation (238/240, the 2 others exceed 16 qubits).
- Nam pf2_10 on Heron is routed over 20 physical qubits, beyond the main run's 16-qubit cap, so it was re-run and verified separately (same totals).
- FakeTorino has 11 dead couplers (error 1.0), removed for *all* pipelines after a first run put Qiskit L3 on them (ESP $\sim 10^{-22}$).

Part VII

# Wrap-up

Lessons, limitations, critique of the original paper, and next steps.

# Lesson, “exact” is easy to get subtly wrong

Found on Nam `pf1_10/pf2_10` (rotations of $\sim 10^{-7}$ rad), whose outputs at first verified only to $10^{-6}$.

## Pitfalls

- `np.allclose` has a default *relative* tolerance $10^{-5}$, so near-commuting gates were swapped.
- $|\mathrm{tr}(U^\dagger V)|/d \approx 1$ is *quadratic* in the error and accepts $\|U - V\| \sim 10^{-4}$.
- Native gates reach only a *discrete* set of global phases.

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## Fixes

- Linear operator-distance tests, one tolerance $10^{-10}$, `rtol=0`.
- Replacement phases snapped to the reachable set, with residuals rejected or charged to $\varepsilon$.
- On ions $\pm i$ is absorbed into an existing MS gate.

**Takeaway.** `pf1_10` now verifies to $5 \cdot 10^{-13}$ (was $1.8 \cdot 10^{-7}$). The original paper uses three different tolerances up to $10^{-3}$.

# Limitations of my work

- **Noise models, not hardware.** ESP and TVD come from calibration snapshots, with gate errors only (no readout, idle or crosstalk). Ion parameters are assumptions.
- **Global phase is not free for us.** Strict mode costs 240 single-qubit gates over the Heron suite. With phase as metadata (as Qiskit, TKET and BQSKit do), all gates $\times 0.838 \rightarrow \times 0.825$.

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- **Global phase is not free for us.** Strict mode costs 240 single-qubit gates over the Heron suite. With phase as metadata (as Qiskit, TKET and BQSKit do), all gates $\times 0.838 \rightarrow \times 0.825$.
- **Baselines.** BQSKit (approximate) needs fewer 2q gates than we do. Quartz/QUESO were not run, and Qiskit, TKET and BQSKit use one seed.
- **Local only.** Layout and routing are Qiskit's, windows have $\leq 3$ qubits, and the code is single-threaded Python (median 10 s, max 180 s per circuit vs. $<$1 s for Qiskit).
- **Scope.** Circuits up to $\sim$1,000 gates, and no mid-circuit measurement or feed-forward.

# Weaknesses of the original paper

1. **Missing structure.** Clifford gate sets, but no tableaux and no exact keys, floating-point lookup with three tolerances, and a classifier to speed up an avoidable $O(N)$ scan.
2. **No notion of optimality.** Only compared with Qiskit, and 4.5$\times$ from an optimum computable in minutes.
3. **Equivalence.** Up to phase only (unsound under control), with no state/measurement context and no error tracking.
4. **Hardware model.** Discrete angles, all-to-all, total count. Matches neither IBM nor ion hardware.

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5. **Evaluation.** Random benchmarks mix, so "from 300 gates" measures input length. One circuit per table, and MATLAB vs. compiled timings.
6. **Search.** Contiguous windows, random swaps and no convergence criterion. Radius 5 is the binding limit.
7. **V3.** Unreported features and threshold, 30 % recall at default.
8. **Hardware evidence.** One 2-qubit circuit and no error bars. The effect vanishes at Qiskit level 3.
9. **Reproducibility.** Only V2 released, with visible bugs, and the database for Tables III–V unstated.

## About the paper

- Reproduces, with qualifications on database, forest threshold, baselines and transpiler level.
- Its gates are Clifford, and exact optima show it is 3–4.5$\times$ from optimal.
- Theory explains why. Radius 5 and window 8 sit below the exactness threshold, and random benchmarks measure a constant.

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## About the extension

- A general optimiser built on "equivalence = group action", with soundness and error bounds.
- Proven optimum on 35/40 of the paper's 3-qubit circuits without Clifford knowledge.
- On 58 real algorithms, stacked on Qiskit L3, it never adds 2q gates, never lowers ESP and removes 13 % of gates on Heron.
- BQSKit's approximate resynthesis needs fewer 2q gates. Exact rewriting keeps the unitary, removes more single-qubit gates and is faster. Run after BQSKit, it removes a further 16 % of gates on

# Next steps, and questions for you

## Next steps

1. Layout-aware rewriting with calibrated costs (the biggest remaining lever on Heron).
2. Cheaper exact phase repair on Heron.
3. 4-qubit windows via symmetry reduction, or a learned admissible A$^*$ heuristic trained on exact tables.
4. Ground truth beyond Clifford, with exact Clifford+T normal forms for small $T$-counts.
5. Reach BQSKit's 2q counts without approximation (larger windows, snapped resynthesis), a Quartz baseline, and real-hardware runs with error bars.

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## Where I'd like your input

1. Is the target a short note on the reproduction + optimality gap, or a full paper on the general framework?
2. Is real-hardware access available for an error-barred experiment?
3. Which venue fits, a quantum-computing one (theory + benchmarks) or a compiler venue?
4. Should we contact the authors about the unstated database and the code bugs?

# Deliverables

| What | Where |
|---|---|
| Study notes (read instead of the paper) | `manuscripts/notes/notes.pdf` |
| Reproduction report (every table and figure) | `manuscripts/reproduction/reproduction_report.pdf` |
| Extension paper | `manuscripts/extension/extension_paper.pdf` |
| Python port, exact Clifford engine, general optimiser | `qcr/` |
| Experiment scripts and raw results | `experiments/`, `results/` |
| This talk | `manuscripts/talk/talk.pdf` |

## Thank you

# Backup

# Backup. Paper-benchmark numbers (mean total gate count)

| set | Qiskit L3 | paper's method | Clifford CW-sem | optimum / bound | general |
|---|---|---|---|---|---|
| T3 (ion, 3q, 20) | 127.2 | 47.1 | – | 10.4 | **10.8** |
| T4 (IBM-style, 3q, 20) | 104.0 | 36.8 | – | 12.2 | **12.5** |
| journal ion (4q, 100) | 164.9* | 78.8* | 45.6 | $\geq$11.1 | **46.1** |
| journal IBM-style (4q, 100) | 140.9* | 76.0* | 49.3 | $\geq$12.9 | **51.4** |
| T5 (IBM-style, 15q, 10) | 332.1 | 335.9 | 324.6 | – | **327.7** |

*first 30 circuits. The general engine uses up-to-phase semantics (as in the paper), total count, discrete $\pm\pi/2$ alphabets and all-to-all connectivity.

# Backup. Minimal entangling-gate counts, 3 qubits

| entangling gates | 0 | 1 | 2 | 3 | 4 | 5 | 6 |
|---|---|---|---|---|---|---|---|
| 3-qubit Cliffords | 13,824 | 373,248 | 5,971,968 | 38,486,016 | 42,052,608 | 5,971,968 | 27,648 |

Mean 3.51, maximum 6. Identical for both alphabets ($R_{XX}(\frac{\pi}{2})$ and CZ are locally equivalent). For 2 qubits the counts are 576 / 5,184 / 5,184 / 576 for 0–3, the known CNOT-count distribution.

# Backup. Worked examples from the reproduction

- **Fig. 1.** The 8-gate and 4-gate Bell circuits both equal CNOT $(H \otimes I)$ to $10^{-13}$, and no 3-gate circuit exists.
- **Fig. 4 analogue** (2 qubits, 40 gates). $40 \to 34 \to 26 \to 24 \to 20 \to 18 \to 16 \to 8 \to 4$, and every step removes an even number of gates (parity theorem). The paper's final circuit contains an inverse pair, and the optimum is 2.
- **Fig. 5 analogue** (3 qubits, 200 gates). $200 \to 36$ in 31 replacements.
- **Authors' demo circuit** (5 qubits, 300 gates). Authors 140, our port 140 (132–148) with the demo database and 115 (112–117) with the 24-op depth-5 database.